Biliary Atresia, Part One: Epidemiology and Phenotypes. Isolated Disease, Splenic Malformation, Cytomegalovirus-Associated Disease and Cystic Atresia.

Picture three infants who all arrive at the same surgical list in the same month. The first is a six-week-old girl, born at term, whose stools turned pale at three weeks. She has no other anomaly. The second is a five-week-old boy with a heart on the right side of his chest, a liver that lies across the midline, and a mother who has had type one diabetes since childhood. The third had a small cyst at the hilum of the liver on her twenty-week anomaly scan, was thought to have a choledochal cyst, and at two weeks of age already has pale stools. All three will be given the same diagnosis, biliary atresia, and all three will be offered the same operation, the Kasai portoenterostomy. Yet they almost certainly have different diseases that began at different times, and they will not have the same outcome.

The clinical question of this chapter is simple to ask. How common is biliary atresia, where and in whom does it occur, and does it matter which kind of biliary atresia an infant has? The main idea is that biliary atresia is a final common pathway rather than a single disease. It is a fibro-obliterative cholangiopathy, meaning a process that inflames, scars and finally closes the bile ducts outside the liver, and it extends into the ducts inside the liver as well. Different routes lead to that same end point, and they start at different moments: some in the first weeks of embryonic life, some in the second trimester, and some perhaps around birth. The incidence of the disease varies more than tenfold around the world. And the phenotype, the clinical pattern the infant shows, tells you when the injury probably began and how the Kasai operation is likely to go.

A note on sources. The diagnostic framework is the 2017 joint guideline on cholestatic jaundice in infants written by Fawaz and colleagues for the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN. The phenotype framework comes mainly from Mark Davenport and the King's College Hospital group in London, in a 2006 study of the splenic malformation syndrome, a 2021 review of clinical phenotypes, and a 2025 review of syndromic variants. Population data come from England and Wales, published by Livesey and colleagues in 2009; from the United States, published by Hopkins and colleagues in 2017; from the Texas Birth Defects Registry, published by Cavallo and colleagues in 2022; from Sweden, published by Fischler and colleagues in 2002; from Taiwan in 2011; from Saudi Arabia in 2022; and from the Japanese Biliary Atresia Registry, summarised by Okubo, Nio and colleagues in 2025. The cytomegalovirus data come from a 2021 meta-analysis in Frontiers in Pediatrics. Neither the American Association for the Study of Liver Diseases, known as AASLD, nor the European

Association for the Study of the Liver, known as EASL, has a guideline on the epidemiology or phenotypes of biliary atresia.

Part one. Two clocks: the embryo and the newborn.

To understand the phenotypes, start with how the bile ducts are built. The extrahepatic bile duct, the duct that runs from the liver to the duodenum, forms very early, during roughly the fourth to the sixth week of gestation. Embryologists describe this period as Carnegie stages eleven to seventeen. That is exactly the same window in which the embryo decides its left and right sides, which is called laterality. Laterality determines where the heart points, which side the stomach and spleen lie on, how the great veins drain and how the gut rotates. If something disturbs development in this window, it can damage the bile duct and the laterality programme at the same time. That is the simplest explanation for the splenic malformation form of biliary atresia, in which the infant has both atresia and laterality defects.

Next, at about twelve weeks of gestation, the extrahepatic duct joins the network of ducts that is forming inside the liver at the hilum. Injury after this point would be expected to spare laterality. Cystic biliary atresia is thought to begin after the twelfth week. The remnant of the duct dilates above a segment that has closed, and by sixteen to twenty weeks this cyst can be seen on the routine anomaly scan.

The third observation moves the story to birth. In 2011, Harpavat, Finegold and Karpen at Texas Children's Hospital looked back at newborn blood tests from infants who were later diagnosed with biliary atresia. The direct or conjugated bilirubin was already abnormally raised in the first one or two days of life, long before anyone noticed jaundice or pale stools. The conclusion is that, in most infants, biliary atresia is already present at birth. It is not a disease that starts at three or four weeks of age, even though that is when we usually see it.

Finally, in some infants, there may be a perinatal trigger. The candidate with the most evidence is cytomegalovirus, usually shortened to CMV, the common herpes virus that infects about one to two percent of all newborns. In these infants, obliteration of the ducts continues over the first weeks, the stools turn pale, and the window for the Kasai operation opens and begins to close.

Why do these two clocks matter in clinic? Because the earlier the injury began, the less any treatment given after birth, such as corticosteroids or an antiviral drug, can be expected to change the course. In an embryonic disease, the outcome depends mostly on the timing of surgery and on the associated anomalies, especially the heart.

Part two. How common is biliary atresia, and where?

Start with the bigger picture. The joint NASPGHAN and ESPGHAN guideline states that cholestatic jaundice in infancy affects about one in every two thousand five hundred term infants, and that biliary atresia accounts for twenty-five to forty percent of those cases. It is

therefore the single most important cause of conjugated jaundice in the first months of life.

The incidence of biliary atresia itself varies widely. The deck draws this as a horizontal bar chart of cases per ten thousand live births, and the bars fall from the Pacific to the Middle East. The highest reported figure comes from French Polynesia, at about three point two per ten thousand, which is roughly one in three thousand one hundred births. Among New Zealand infants of Maori ancestry the figure is about one in five thousand three hundred, compared with about one in sixteen thousand among infants of European ancestry in the same country. In Taiwan, national data for 1996 to 2008 gave one point four eight per ten thousand, or about one in six thousand eight hundred. The guideline quotes one in six thousand for Taiwan, and East Asian figures generally lie between one in five thousand and one in ten thousand.

Europe sits much lower. Sweden, in a national register study of one point two million children, had eighty-five cases, or about one in fourteen thousand. England and Wales, in a prospective national database of three hundred and two infants, had zero point five eight per ten thousand, which is about one in seventeen thousand. France and the Netherlands both lie at about zero point five per ten thousand, around one in nineteen thousand to one in twenty thousand. In the United States, the Texas Birth Defects Registry found zero point six five per ten thousand, while a national analysis of hospital billing data by Hopkins and colleagues found about one in twenty-two thousand. The lowest figure in this set comes from Saudi Arabia, with about one in forty-four thousand.

Two lessons follow. First, the range from Saudi Arabia to French Polynesia is about fourteen-fold, and Asia and the Pacific sit highest. Second, the method changes the number. Registries that actively look for every case find more than studies that rely on billing codes, so the United States figure of one in twenty-two thousand from billing data and the Texas registry figure are not really in conflict. The joint guideline's own numbers, one in twelve thousand for the United States, one in nineteen thousand for Canada and one in eighteen thousand for Europe, are reasonable summaries but should be read with that caveat.

Part three. Inside one country, the risk is not even.

The differences are not only between countries. In England, Livesey and colleagues found an incidence of about zero point three eight per ten thousand in the north-west and zero point seven eight per ten thousand in the south-east, an odds ratio of about two. Davenport has pointed out that London, where about forty percent of births are to non-white mothers, has an incidence of about one in fourteen thousand, while the north-west of England, with about fifteen percent non-white births, has about one in twenty-two thousand seven hundred.

In the United States, the billing data showed a relative risk of about one point nine for Asian and Pacific Islander infants and about one point three for Black infants, compared with the reference group. In the Texas registry, infants of Black mothers had an adjusted prevalence

ratio of about one point five compared with infants of White mothers. And in England, South Asian infants reached the Kasai later, at a median of fifty-nine days compared with fifty-two days, which is a reminder that ancestry can change access as well as risk.

Girls are affected more often than boys. The relative risk for female sex was about one point four in the national United States data, the adjusted prevalence ratio was about one point seven in Texas, and about two-thirds of infants in the Japanese registry are female.

How much of this is genetic? Probably not much. Variants in genes that control the cilia, the tiny hair-like structures on the cells lining the bile duct, have been linked to biliary atresia, but they imply only about a two-fold increase in risk. Environment and development seem to matter more. The practical point is that ancestry shifts the prior probability, but it never removes biliary atresia from the differential diagnosis of an infant with pale stools.

Part four. Risk factors, and a season that did not hold up.

The Texas registry study by Cavallo and colleagues is the best modern look at risk factors. It compared three hundred and five cases with nearly four point seven million births between 1999 and 2014. The deck shows its results as a forest plot of adjusted prevalence ratios on a logarithmic scale, where a value of one means no difference. Birth before thirty-two weeks of gestation had the largest effect, with a ratio of three point eight five, and a confidence interval from about two point four to six point two. Maternal diabetes came next, at two point three four, with an interval from about one point six to three point five, and the association was stronger for diabetes that was present before pregnancy than for gestational diabetes. Female sex had a ratio of one point six eight, birth at thirty-two to thirty-seven weeks one point six four, and a Black mother one point five four. These associations held even when the analysis was limited to isolated biliary atresia without other birth defects, which supports the idea that events in the womb shape even the commonest form of the disease.

The Swedish study found a similar picture: older maternal age, with an odds ratio of about three; four or more previous births, with an odds ratio of about two; prematurity, with an odds ratio of about three; and being small for gestational age, with an odds ratio of nearly five. Davenport's 2025 review adds a striking figure from the Texas registry: among infants with laterality defects, pregestational maternal diabetes carried an odds ratio of about twenty-six. That fits the embryonic story of the splenic malformation form.

What about the season? Small regional series from Texas and Atlanta in the 1980s and 1990s reported clusters of births, in autumn in one report and in winter in another, and that was taken as a hint of a viral cause. Larger studies have not reproduced it. A Michigan study of sixty-five infants, with enough power to detect a thirty percent swing in birth rates over three months, found no monthly or seasonal variation in either births or conceptions. The Swedish national study found no difference by birth month. England and Wales found September to be the commonest birth month and December the rarest, but the difference was not significant,

with a p value of zero point two. Saudi Arabia found no seasonal clustering, although there were three peak years. The Texas registry found only a weak signal, with autumn conceptions somewhat less likely than spring conceptions. The fair conclusion is that a viral trigger may still matter in a subgroup, but seasonality is not a clinical tool, and it should never be used to make the diagnosis more or less likely.

Part five. Three groups in the guideline, five in the clinic.

The joint guideline divides biliary atresia into three groups according to associated anomalies. The non-syndromic form, in which biliary atresia is the only anomaly, accounts for about eighty-four percent. About six percent have at least one other malformation but no laterality defect. And about ten percent have laterality defects, usually with an abnormal spleen. This last group is the splenic malformation syndrome.

The King's College group uses five clinical phenotypes. Isolated biliary atresia makes up about two-thirds to nine-tenths of series. The biliary atresia splenic malformation syndrome, usually shortened to BASM, makes up about ten to fifteen percent of Western series, fewer than three percent in Japan and fewer than one percent in China. Cystic biliary atresia makes up somewhere between five and twenty-two percent, depending on how a cyst is defined. Cytomegalovirus IgM-positive biliary atresia, in which the infant has antibodies of the immunoglobulin M class against cytomegalovirus at diagnosis, makes up about five to ten percent in Europe and more in China. And a final group of other syndromic forms includes the cat-eye syndrome, cardiac-associated atresia and atresia with duodenal atresia.

These two systems are not interchangeable. The guideline groups infants by anatomy and associated anomalies, while the phenotypes add timing and cause. Cytomegalovirus-associated and cystic disease are not classes in the guideline at all, yet they carry some of the strongest prognostic signals. In practice, the phenotype should be written in the operative note and in any registry, because it changes counselling and because it is the only way to compare outcomes fairly between centres.

There is also a surgical classification by the level of the blockage, and the Japanese registry gives the best numbers, from three thousand nine hundred and eleven infants. Type one, in which the common bile duct is blocked but the ducts above it are patent, made up about three percent, and a further eight percent were type one with a cyst in the remnant. Type two, with the block at the common hepatic duct, made up under two percent. Type three, with no patent duct at the porta hepatis, the fan of tissue where the ducts leave the liver, made up nearly eighty-seven percent. Clearance of jaundice after the Kasai followed the anatomy: about seventy percent in type one, seventy-nine percent in type one with a cyst, seventy-two percent in type two, and sixty percent in type three. The same registry shows a mean age at the Kasai of about sixty-five days, falling from about seventy-one days in 2011 to about sixty-one days in 2023, while overall clearance rose from about fifty-eight percent to about sixty-four percent. And almost nine percent of Japanese infants had a bleed from vitamin K deficiency

before their Kasai, which tells us how long cholestasis has usually been present by the time of surgery.

Part six. The splenic malformation syndrome.

The key paper is Davenport's 2006 study of five hundred and forty-eight infants treated at King's College Hospital between 1977 and 2004. Fifty-six of them, about ten percent, had the splenic malformation syndrome, and they were more often female. About thirty-seven percent had situs inversus, meaning a mirror-image arrangement of the organs, and about forty-five percent had a cardiac anomaly. Maternal diabetes was present in twelve and a half percent of mothers of these infants, compared with about one percent of mothers in the comparison group. Native-liver survival, which means survival without a liver transplant, was forty-six percent at five years and thirty-two percent at ten years. Interestingly, liver histology, including the degree of fibrosis, and the tissue type did not differ from other infants with atresia.

When you suspect this syndrome, look for polysplenia, meaning several small spleens, or asplenia, meaning no spleen; a portal vein that runs in front of the duodenum, called a preduodenal portal vein; an absent segment of the inferior vena cava within the liver; malrotation of the gut; and situs inversus. These matter for more than the diagnosis. They make the Kasai technically harder, the cardiac lesion may dominate the infant's course, and the vascular anatomy changes how a later transplant is planned.

The syndrome also has a geography. In the deck this is drawn as a bar chart of the share of all biliary atresia. King's College had about ten percent, a Denver series about twelve percent, and Saudi Arabia about eleven percent. Japan had about two percent, and a Chinese series of eight hundred and fifty-one infants had only about half of one percent with polysplenia. Yet when the Japanese registry expresses the syndrome as cases per ten thousand births, at about zero point zero two three, it is in the same range as Europe and the United States, at about zero point zero three to zero point zero eight. In other words, the high incidence of biliary atresia in East Asia is an excess of isolated disease, not of the embryonic splenic malformation form.

Genetics has begun to explain part of the syndrome. In a North American sequencing study of sixty-seven infants, five had two damaging variants in a ciliary gene called PKD1L1. Only sixty percent of those five had a splenic malformation, which shows that the genotype and the clinical phenotype overlap but are not the same thing.

Part seven. Cytomegalovirus and biliary atresia.

The cytomegalovirus-associated phenotype is the most debated. In a United Kingdom series about ten percent of infants with atresia were CMV IgM-positive, and Chinese series report up to thirty percent or more. In Denver, cytomegalovirus-specific T-cell responses were found in fifty-five percent of infants at diagnosis. Infants in this group tend to be older at diagnosis,

to have a larger spleen, and to show more inflammation and fibrosis in the liver.

The 2021 meta-analysis pooled nine studies with six hundred and ninety-four infants, of whom two hundred and ninety-four were cytomegalovirus-positive. The deck shows the result as a forest plot. Cytomegalovirus-positive infants had about half the odds of clearing jaundice, with an odds ratio of zero point four seven and a confidence interval from about zero point three to zero point seven, and the studies agreed well with each other. The effect was clearer for clearance within six months, with an odds ratio of zero point three one, than for clearance within three months, where the interval included one. For native-liver survival at two years, the hazard ratio was about one point six, but the confidence interval ran from about zero point seven to about three point eight, so the effect on survival is not proven. Cholangitis, meaning infection of the bile ducts after the Kasai, looked more common, but that rested on only two studies and was not significant. One more caution: forty-two percent of infants in these studies were cytomegalovirus-positive, far above what we would expect from population rates, which tells us that the studies selected and defined their patients in different ways.

Here the guideline and the phenotype literature part company, and this is the most important disagreement of the topic. The joint guideline says that a causal role for cytomegalovirus in biliary atresia remains unproven. It recommends urine culture or polymerase chain reaction, usually called PCR, to diagnose the infection, and it calls cytomegalovirus IgM of limited value and possibly less sensitive. The King's College group, by contrast, defines its prognostic subgroup by serum IgM. A further difficulty is that urine PCR after the first three weeks of life cannot tell congenital infection from infection acquired during or after birth, for example from breast milk.

What about treatment? The King's group reported in 2019 that giving intravenous ganciclovir or oral valganciclovir after the Kasai, in selected cytomegalovirus IgM-positive infants, was followed by better outcomes than in earlier untreated infants. They called this prima facie evidence, which is an honest description: it was a single-centre, non-randomised comparison with historical controls. No guideline takes a position.

A sensible stance is to send serum cytomegalovirus IgM and a blood or urine PCR at presentation in every infant with suspected atresia, to record the result clearly, and to discuss antiviral treatment only within a protocol or a trial. And never let a positive cytomegalovirus result stop the work-up. Cytomegalovirus can coexist with atresia, so an infant with pale stools needs a cholangiogram regardless of the virology.

Part eight. Cystic biliary atresia, a gift of time.

Cystic biliary atresia is the phenotype most likely to be recognised before birth, because a cyst at the hilum can be seen on the eighteen to twenty week scan. In the Japanese registry, about five and a half percent of all infants with atresia were diagnosed antenatally. The

danger is to mistake it for a choledochal cyst, a congenital dilatation of the bile duct that is usually repaired electively. The clues to atresia are a cyst that is usually smaller, a small or irregular gallbladder, and no dilatation of the ducts inside the liver. But the real decision rests on the stool colour and on the conjugated bilirubin, not on the cyst itself. If the stools are pale and the conjugated bilirubin is raised, the cyst should be treated as cystic atresia and the infant needs a cholangiogram and a Kasai on the atresia timetable, not an elective operation in a few months.

The reward for early recognition is large. Davenport's group reports clearance of jaundice and native-liver preservation in more than three-quarters of infants with cystic atresia after early surgery, and outcome is strongly tied to age at operation. In the Japanese registry, clearance was seventy-nine percent for type one with a cyst, compared with sixty percent for type three. Some of these patients do decompensate in adult life, so they need lifelong follow-up, like every survivor of a Kasai.

Part nine. Other associations, and when the heart sets the outcome.

About six percent of infants have another malformation without a laterality defect. In a King's series of thirty-seven infants with cardiac-associated atresia, about seven percent of the total, almost half had features of the splenic malformation syndrome and about one in ten had the cat-eye syndrome. Their outcomes were worse for reasons that had little to do with the liver: overall mortality was forty percent compared with four percent in contemporary controls, and native-liver survival at five years was fifty-four percent compared with sixty-five percent. Duodenal atresia is another clue: in ten of thirteen infants who had both duodenal and biliary atresia, the splenic malformation syndrome was also present.

The cat-eye syndrome is a chromosomal disorder involving extra material from the proximal part of chromosome twenty-two, and in one series four of five affected infants had biliary atresia. Kabuki syndrome, Zimmermann–Laband syndrome, Kartagener syndrome, Hirschsprung disease and oesophageal atresia have also been reported with atresia. The practical message is that every infant with biliary atresia should have an echocardiogram before the Kasai, an ultrasound that maps the abdominal veins, and a chromosomal study if the infant looks dysmorphic.

Part ten. What the guidelines say.

Comparing the joint guideline with the United Kingdom phenotype literature gives a mixed picture. On incidence, the guideline's figures differ from newer data, which suggest a lower rate in the United States when billing data are used. On classification, the two differ: the guideline uses three anatomical groups, the clinic uses five phenotypes. On the clues to the splenic malformation syndrome, they are aligned: both point to heterotaxy, polysplenia, a preduodenal portal vein and congenital heart disease. On cystic atresia they are also aligned, with the guideline noting that a cyst may coexist with atresia. On the role of cytomegalovirus,

the difference is material: the guideline calls the causal role unproven, while the phenotype literature treats cytomegalovirus-associated atresia as a distinct cause and prognostic group, defined by a test the guideline considers insensitive. On antiviral treatment, and on ethnicity and season, there is simply a gap: no guideline addresses them. The material gap, in one sentence, is that no guideline tells us how to define, test for or treat cytomegalovirus-associated atresia.

What to do differently in clinic.

Consider a few infants. An infant with an antenatal hilar cyst of one and a half centimetres who has pale stools and a conjugated bilirubin of about two and a half milligrams per decilitre at two weeks should be treated as cystic atresia, with an urgent cholangiogram and Kasai, not booked for elective cyst surgery. A jaundiced five-week-old with dextrocardia and a midline liver should be assumed to have the splenic malformation syndrome until proven otherwise, with an echocardiogram, mapping of the portal vein and inferior vena cava, and surgery planned with cardiology. A seven-week-old with pale stools, a positive serum cytomegalovirus IgM and a spleen four centimetres below the ribs should not be labelled as cytomegalovirus hepatitis and observed; that infant needs a cholangiogram, and if atresia is confirmed, the record should say cytomegalovirus-associated, with antivirals considered only within a protocol. A mother with diabetes who asks whether she caused her baby's disease can be told honestly that diabetes is associated with atresia, with a prevalence ratio of about two, and that association is not the same as cause. A centre whose clearance rate is forty-eight percent should not compare itself with a published sixty percent without first adjusting for its phenotype mix and its age at Kasai. And an infant of South Asian heritage referred at seventy days reminds us that later presentation in this group is documented, so stool colour education needs to reach every community.

So tomorrow in clinic, remember that biliary atresia is about one in seventeen thousand births in England, one in fourteen thousand in Sweden, and one in six to seven thousand in Taiwan, with more girls than boys and higher rates after maternal diabetes and preterm birth. Do not use the season to judge the likelihood. Name the phenotype in every case: isolated, splenic malformation, cytomegalovirus-associated, cystic, or other syndromic. Look for the laterality and cardiac anomalies before surgery, treat a small hilar cyst with pale stools as atresia, and never let a positive cytomegalovirus test delay a cholangiogram. And above all, remember that the disease is already present at birth, so the stool colour and the conjugated bilirubin, not the calendar or the family's ancestry, decide when to act.